

Lists

1. Create a list called `my_list` with the three following items. Name them `"char_vec"`, `"char_mat"`, and `"lst"`. Draw this list as a train with cargo in the space below.

```
list_el_1 <- "crane"
list_el_2 <- matrix(c("a", "b", "c", "d"))
list_el_3 <- list(c(3.2, 5.7, 9), c(FALSE, TRUE))
```

For each of the following, write the R code to subset the list and then draw the corresponding train or cargo. You're welcome to subset by index, name, or logical; and use `[]`, `[[ ]]`, or `$`, as is appropriate.

2. The same list but without `char_mat`.
3. The character string `"crane"`.
4. The element `"d"`.
5. The element `"d"` repeated several times to form a character vector of length 4.
6. The double vector, as the sole element of a list.
7. The value `TRUE`.

Control Flow

8. Write an if statement that takes an exam score like `score <- 93` and returns "A" if the score is greater than 90.
9. Write an if statement that takes a single integer `n` and prints "even" if `n` is divisible by 2, "odd" otherwise. Test your code with `n <- 7` and `n <- 16`. (*Hint 1: experiment with the modulo operator `%%`, e.g `3 %% 2`.)
10. Extend your previous answer to return letters grades of A, B, C, or "no pass" if the score is in the 90s, 80s, 70s, or less than 70, respectively.
11. On the lines provided below, complete the code to iterate through the vector `x` and print its contents. For this question, the iterator, `i`, should take as its values the character string elements of `x`. (*Hint: see Option 3 in the slides*)

```
x <- c("apple", "banana", "cherry")
```

```
for (i in _____) {  
  _____(_____  
}
```

12. Repeat the question but this time, the iterator, `i`, should take as its values the integers from 1 to 3.
(Hint: see Option 1 in the slides)

```
for (i in _____) {  
    _____(  
}  
}
```

13. Write a for loop that uses `x` and prints the following output:

```
Fruit 1: apple  
Fruit 2: banana  
Fruit 3: cherry
```

14. Using a for-loop (don't cheat with vectorization!), change each of the temperatures of `temps_c` one-by-one so that they are converted from degrees Celcius to Fahrenheit.

```
temps_c <- c(0, 10, 20, 30)
```

15. For-loops, like if-else statements, can be *nested*: inside the two for-loops below, your code has access to both of the iterators `i` and `j`. Fill in the code below to turn `m` from a matrix of 0s to a multiplication table showing the products of the integers 1, 2, 3 with one another.

```
m <- matrix(rep(0, 9), nrow = 3)  
  
for (i in _____) {  
    for (j in _____) {  
        m[_____] <- _____  
    }  
}
```

Run the code in R to check your answer. As a cherry on top, make the row and column names of this multiplication table easier to read by reassigning their names with `rownames(m) <- ____` and `colnames(m) <- ____`.

16. *For fun*: Run the following code and view the result.

```
plot(0, 0)

for(i in 1:100) {
  points(runif(1, min=0, max=1), runif(1, min=0, max=1))
}
```

`plot(0, 0)` makes a scatterplot with a single point at (0, 0). `points(x, y)` will add a point to your plot at (x, y) and `runif()` works like `rnorm()` except it generates 1 random uniform number between 0 and 1.

Modify this code so that your random points appear uniformly along the unit circle (a circle centered at (0, 0) with radius one) instead of uniformly in the upper right quadrant.

17. *For even more fun*: Go onto the class JupyterHub to access the Data In Trees files. Write an R script that uses a for loop to read all of the text files it R and forms a list where the first element is a character vector named `song` and the second element is a double vector named `distance`. The element index in each vector should correspond to the same student's index card.

If you figure it out, please share your R script on Ed!

More questions coming soon . . .